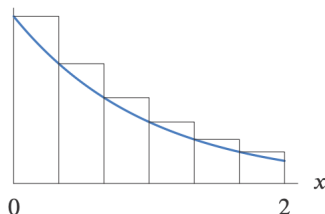


The Definite Integral

Problem 1

Consider the following Riemann sum approximation with n subdivisions to $\int_a^b f(x)dx$.



- (a) Is it a left or right Riemann sum? Would the other one be larger or smaller?
 (b) What are a, b, n and Δx ?

Solution.

- (a) Since the height of each rectangle is the value of the function at the left endpoint in each sub-interval, it is a left Riemann sum. The right Riemann sum would be smaller as the function is decreasing.
 (b) In this case $a = 0$, $b = 2$, $n = 6$ and $\Delta x = \frac{b-a}{n} = \frac{2}{6} = \frac{1}{3}$. ■

Problem 2

Use the following table of values to estimate $\int_0^{40} f(x)dx$.

x	0	10	20	30	40
$f(x)$	350	410	435	450	460

Solution.

Any choice of Riemann sum will give a suitable approximation. For instance, using a left Riemann sum gives

$$\int_0^{40} f(x)dx \approx 10(350 + 410 + 435 + 450) = 16450$$

and using a right Riemann sum yields

$$\int_0^{40} f(x)dx \approx 10(410 + 435 + 450 + 460) = 17550.$$

It should come as no surprise that the right Riemann sum is larger than the left Riemann sum since f is increasing. ■

Problem 3

For $f(x) = 3x + 2$, evaluate the Riemann sum

$$\sum_{i=0}^3 f(x_i) \Delta x,$$

where $\Delta x = 2$ and $x_0 = 3$.

Solution.

Since $x_0 = 3$ and $\Delta x = 2$,

$$\begin{aligned} \sum_{i=0}^3 f(x_i) \Delta x &= \Delta x (f(x_0) + f(x_0 + \Delta x) + f(x_0 + 2\Delta x) + f(x_0 + 3\Delta x)) \\ &= 2(f(3) + f(5) + f(7) + f(9)) \\ &= 2(11 + 17 + 23 + 29) \\ &= 160. \end{aligned}$$

■

Problem 4

Estimate $\int_0^1 e^{-x^2} dx$ using $n = 5$ rectangles and a left Riemann sum.

Solution.

For a left Riemann sum on the interval $[0, 1]$ with $n = 5$ rectangles, we have

$$\Delta x = \frac{1 - 0}{5} = \frac{1}{5}.$$

It follows by definition of a left Riemann sum that

$$\int_0^1 e^{-x^2} dx \approx \sum_{i=0}^4 f\left(\frac{i}{5}\right) \Delta x = \frac{1}{5} \left(1 + e^{-0.2^2} + e^{-0.4^2} + e^{-0.6^2} + e^{-0.8^2}\right) \approx 0.808.$$

■

Problem 5

Evaluate each of the following limits.

(a) $\lim_{n \rightarrow \infty} \sum_{i=1}^n \left(\frac{i}{n}\right)^2 \cdot \frac{1}{n}$

(b) $\lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} \sqrt{4 - \left(\frac{2i}{n}\right)^2} \cdot \frac{2}{n}$

Solution.

(a) If $f(x) = x^2$ and $x_i = \frac{i}{n}$, then $\Delta x = x_{i+1} - x_i = \frac{1}{n}$ and

$$\sum_{i=1}^n \left(\frac{i}{n}\right)^2 \cdot \frac{1}{n} = \sum_{i=1}^n f(x_i) \Delta x.$$

Since $x_0 = 0$ and $x_n = 1$, this right Riemann sum converges to the definite integral

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \left(\frac{i}{n}\right)^2 \cdot \frac{1}{n} = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x = \int_0^1 f(x) dx = \int_0^1 x^2 dx = \frac{1}{3}.$$

(b) If $f(x) = \sqrt{4-x^2}$ and $x_i = \frac{2i}{n}$, then $\Delta x = x_{i+1} - x_i = \frac{2}{n}$ and

$$\sum_{i=0}^{n-1} \sqrt{4 - \left(\frac{2i}{n}\right)^2} \cdot \frac{2}{n} = \sum_{i=0}^{n-1} f(x_i) \Delta x.$$

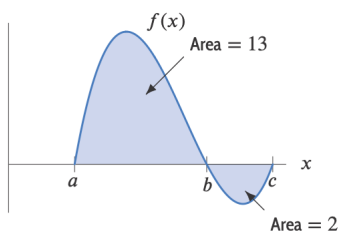
Since $x_0 = 0$ and $x_n = 2$, this left Riemann sum converges to the definite integral

$$\lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} \sqrt{4 - \left(\frac{2i}{n}\right)^2} \cdot \frac{2}{n} = \lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} f(x_i) \Delta x = \int_0^2 f(x) dx = \int_0^2 \sqrt{4-x^2} dx = \frac{1}{4} \cdot \pi \cdot 2^2 = \pi,$$

where we use the fact that the definite integral corresponds to the area in the first quadrant of a circle of radius two. ■

Problem 6

Consider the function f with graph



Evaluate each of the following integrals.

(a) $\int_a^b f(x) dx$

(c) $\int_a^c f(x) dx$

(b) $\int_b^c f(x) dx$

(d) $\int_a^c |f(x)| dx$

Solution.

(a) The definite integral gives the net area and f is positive on $[a, b]$. This means that

$$\int_a^b f(x) dx = 13.$$

(b) The definite integral gives the net area and f is negative on $[b, c]$. This means that

$$\int_b^c f(x) dx = -2.$$

(c) Since the definite integral gives the net area,

$$\int_a^c f(x) dx = 13 - 2 = 11.$$

(d) Integrating the absolute value of a function gives the total area as opposed to the net area. This means that

$$\int_a^c |f(x)| dx = 13 + 2 = 15. \quad \blacksquare$$

Problem 7

Without evaluating them, decide which of the two definite integrals in each pair is smaller.

(a) $\int_0^1 x dx$ and $\int_0^1 x^2 dx$

(b) $\int_2^3 \cos(x) dx$ and $\int_2^3 x dx$

Solution.

(a) Since $x^2 \leq x$ for $x \in [0, 1]$, the monotonicity of the integral implies that

$$\int_0^1 x^2 dx \leq \int_0^1 x dx.$$

(b) Since $\cos(x) \leq 1 \leq 2 \leq x$ for $x \in [2, 3]$, the monotonicity of the integral implies that

$$\int_2^3 \cos(x) dx \leq \int_2^3 x dx. \quad \blacksquare$$

Problem 8

Given that $\int_1^5 f(x) dx = 5$, $\int_1^9 f(x) dx = 3$ and $\int_3^9 f(x) dx = 4$ determine each of the following.

(a) $\int_5^9 f(x) dx$

(b) $\int_3^5 f(x) dx$

(c) $\int_3^1 f(x) dx$

Solution.

(a) The basic properties of the integral imply that

$$\int_5^9 f(x) dx = \int_1^9 f(x) dx - \int_1^5 f(x) dx = 3 - 5 = -2.$$

(b) The basic properties of the integral and part (a) imply that

$$\int_3^5 f(x) dx = \int_3^9 f(x) dx - \int_5^9 f(x) dx = 4 - (-2) = 6.$$

(c) The basic properties of the integral imply that

$$\int_3^1 f(x) dx = - \int_1^3 f(x) dx = - \left(\int_1^9 f(x) dx - \int_3^9 f(x) dx \right) = -(3 - 4) = 1. \quad \blacksquare$$

The Fundamental Theorems of Calculus

Problem 1

Suppose that $F'(x) = f(x)$ and $\int_0^\pi f(x) = -2$.

(a) Is it possible that $F(x) = \cos(x)$?

(b) Is it possible that $F(x) = \sin(x)$?

Solution.

The idea is to check whether or not the First Fundamental Theorem of Calculus is satisfied. In other words, whether or not the candidate for F is such that

$$F(\pi) - F(0) = \int_0^\pi f(x) dx = -2.$$

(a) Since $\cos(\pi) = -1$ and $\cos(0) = 1$, we have $F(\pi) - F(0) = -2$. It is therefore possible that $F(x) = \cos(x)$.

(b) Since $\sin(\pi) = 0$ and $\sin(0) = 0$, we have $F(\pi) - F(0) = 0$. It is therefore not possible that $F(x) = \sin(x)$. ■

Problem 2

Suppose that $\int_3^5 F'(t) dt = 5$ and $F(3) = 2$.

(a) If possible, determine $F(5)$.

(b) If possible, determine $F(4)$.

Solution.

The idea is once again to use the first fundamental theorem of calculus.

(a) By the first fundamental theorem of calculus,

$$F(5) = F(3) + \int_3^5 F'(t) dt = 2 + 5 = 7.$$

(b) It is still true that by the first fundamental theorem of calculus,

$$F(4) = F(3) + \int_3^4 F'(t) dt,$$

however we have no way of determining this integral. It is therefore impossible to evaluate $F(4)$. In fact, it is quite easy to come up with two functions F_1 and F_2 such that $F_1(3) = F_2(3) = 2$ and $\int_3^5 F_1'(t) dt = \int_3^5 F_2'(t) dt = 5$ but $F_1(4) \neq F_2(4)$. ■

Problem 3

The secant function is defined by $\sec(x) = \frac{1}{\cos(x)}$.

- (a) Compute the derivative of $f(x) = \tan(x)$.
- (b) Deduce the value of the integral $\int_{-\pi/4}^{\pi/4} \sec^2(x) dx$.

Solution.

- (a) Recall that

$$\tan(x) = \frac{\sin(x)}{\cos(x)} = \sin(x) \cos(x)^{-1}.$$

A combination of the product rule and the chain rule shows that

$$\frac{d}{dx}(\tan(x)) = \cos(x) \cdot \cos(x)^{-1} - \sin(x) \cdot \cos(x)^{-2} \cdot (-\sin(x)) = 1 + \frac{\sin^2(x)}{\cos^2(x)} = \sec^2(x),$$

where we used that $\sin^2(x) + \cos^2(x) = 1$.

- (b) In (a) we have shown that $\tan(x)$ is an antiderivative of $\sec^2(x)$. It follows by the first fundamental theorem of calculus that

$$\int_{-\pi/4}^{\pi/4} \sec^2(x) dx = [\tan(x)]_{-\pi/4}^{\pi/4} = 1 - (-1) = 2. \quad \blacksquare$$

Problem 4

A company produces coal. Suppose that by 1997 the company has produced a total of 10 billion tons of coal since it was founded, and assume that its rate of annual coal production in billions of tons per year is given by the following table of values.

Year	1997	1999	2001	2003	2005	2007	2009
Rate	1.090	1.094	1.121	1.072	1.132	1.147	1.073

Estimate the total amount of coal produced by the company between its foundation and 2009.

Solution.

Denote by $P(t)$ the total amount of coal in billions of tons produced by the company between its foundation and year t . For instance, $P(1997) = 10$. Implicitly we assume that t is after the year in which the company was founded. By the first fundamental theorem of calculus,

$$P(2009) = P(1997) + \int_{1997}^{2009} P'(t) dt.$$

To estimate the right hand side of this expression from the table of values, we approximate the integral by a left Riemann sum. Notice that $\Delta t = 2$ so

$$P(2009) = P(1997) + 2(1.090 + 1.094 + 1.121 + 1.072 + 1.132 + 1.147) = 23.312.$$

By the year 2009 the company has therefore produced 23.3 billion tons of coal. \blacksquare

Problem 5

Compute the derivative of $f(x) = \int_{e^x}^{\pi} \cos(2t^2) dt$.

Solution.

It would be tempting to say that by the Second Fundamental Theorem of Calculus, the derivative is simply $-\cos(2e^{2x})$; however, this would be incorrect as the second fundamental theorem of calculus does not allow for the limit of integration to depend on any function $a(x)$ other than $a(x) = x$. Nonetheless, we can leverage the Second Fundamental Theorem of Calculus to compute $f'(x)$. To do so, we introduce the anti-derivative F of $\cos(2t^2)$,

$$F(x) = \int_0^x \cos(2t^2) dt.$$

In general we would introduce the anti-derivative of the integrand defining the integral function whose derivative we seek to calculate. The Second Fundamental Theorem of Calculus ensures that $F'(x) = \cos(2x^2)$ since the limit of integration is of the form $a(x) = x$ as required by this result. Moreover, the First Fundamental Theorem of Calculus implies that

$$f(x) = \int_{e^x}^{\pi} F'(t) dt = F(\pi) - F(e^x).$$

It follows by the chain rule that

$$f'(x) = -F'(e^x)e^x = -\cos(2e^{2x})e^x,$$

where we have used the fact that $F(2\pi)$ is a constant to say that its derivative is zero. ■

Problem 6

Determine the tangent line approximation of the integral function

$$f(x) = \int_0^x \frac{\sin(t)}{t} dt$$

about the point $x = 0$.

Solution.

To construct the tangent line approximation of f about the origin, we require the value of $f(0)$ and $f'(0)$. Since the integral of a continuous function over a single point is always null,

$$f(0) = \int_0^0 \frac{\sin(t)}{t} dt = 0.$$

On the other hand, the Second Fundamental Theorem of Calculus implies that

$$f'(x) = \frac{d}{dx} \left(\int_0^x \frac{\sin(t)}{t} dt \right) = \frac{\sin(x)}{x}.$$

Using l'Hopital or a Taylor approximation to let x tend to zero reveals that $f'(0) = 1$. The tangent line approximation of $f(x)$ about $x = 0$ is therefore $P_1(x) = x$. ■